

Efficient Resource Allocation in iRAN SDN Slicing Orchestration

Abstract—The rapid evolution of 6G wireless networks introduces a wide range of heterogeneous services, such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC), each imposing stringent and diverse Quality of Service (QoS) requirements. Efficiently managing scarce and dynamic network resources to meet these demands remains a critical challenge, especially in complex and dense network environments. This paper addresses the problem of intelligent and adaptive resource allocation in 6G Intelligent Radio Access Networks (iRAN), focusing on optimizing task execution across multiple computing platforms. We propose a novel framework that dynamically selects among local gNodeB processing, Software-Defined Networking (SDN) enabled task rerouting, and Multi-access Edge Computing (MEC) offloading to balance latency, reliability, and resource utilization. To effectively navigate the complexity of heterogeneous service classes and dynamic network states, our approach integrates Deep Reinforcement Learning (DRL) based on Proximal Policy Optimization (PPO) with Graph Neural Networks (GNN), enabling context-aware and service-aware decision-making. An analytical latency model supports the DRL agent by quantifying trade-offs across execution modes. Extensive simulations demonstrate that the proposed iRAN framework significantly improves latency performance and QoS compliance while ensuring scalable and energy-efficient management of next-generation wireless networks.

Index Terms—6G, Intelligent RAN, SDN, MEC, Deep Reinforcement Learning, Proximal Policy Optimization, Graph Neural Networks, Quality of Service, Multi-service scheduling.

I. INTRODUCTION

The sixth generation (6G) of wireless networks is poised to revolutionize connectivity by supporting a broad spectrum of heterogeneous services. These include enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), massive Machine-Type Communications (mMTC), as well as emerging service categories such as mobile URLLC (mURLLC) and mobility-enhanced reliable low-latency communication (mbRLLC) [14]. Each service class imposes stringent and often conflicting requirements on latency, reliability, scalability, and mobility, presenting significant challenges to traditional radio resource management paradigms [13].

To effectively meet these demands, recent research in 6G emphasizes the necessity for intelligent, context-aware, and adaptive resource allocation mechanisms capable of managing service diversity in highly dynamic and resource-constrained environments [5]. Architecturally, centralized control via Software-Defined Networking (SDN) [12] and proximity-based computing through Multi-access Edge Computing (MEC) [14] have emerged as pivotal enablers, providing

flexible network programmability and low-latency computation close to the user.

Nevertheless, the inherent spatial-temporal complexity and variability of 6G wireless ecosystems demand more sophisticated solutions. Learning-based methods, particularly Deep Reinforcement Learning (DRL), offer powerful tools for real-time, service-aware resource management by continuously adapting to environmental changes and user demands. In this work, we propose an Intelligent Radio Access Network (iRAN) framework that synergizes DRL, specifically leveraging the Proximal Policy Optimization (PPO) algorithm [15], with Graph Neural Networks (GNNs) [16]. The GNN effectively encodes the underlying network topology and contextual relationships among heterogeneous entities—user equipments (UEs), gNodeBs, and MEC servers—modeled as nodes in a graph linked by wireless and interference-aware edges. This structured embedding guides the DRL agent to make adaptive, context-sensitive resource allocation decisions.

Complementing the intelligent learning framework, we develop a rigorous analytical model that quantifies latency across three fundamental task execution modes: local processing at the serving gNodeB, SDN-facilitated rerouting to neighboring gNodeBs, and offloading to MEC servers. The model incorporates strict Quality of Service (QoS) constraints, enabling the hybrid decision-making process to optimize latency, energy consumption, and service compliance across diverse 6G vehicular and mobile applications.

The rest of this paper is organized as follows: Section II reviews state-of-the-art 6G access technologies and intelligent resource allocation strategies. Section III presents the motivation behind our work, the proposed iRAN system model, and the latency analysis. Section IV describes the proposed approach. Section V provides simulation results and performance evaluation. Finally, Section VI concludes with key insights and future research directions.

II. RELATED WORK

Recent research in 6G Radio Access Networks (RANs) has increasingly adopted Deep Reinforcement Learning (DRL) and Graph Neural Networks (GNNs) to enable intelligent and adaptive resource management. Rezazadeh et al. [1] proposed an explainable DRL framework for 6G RAN slicing, improving policy interpretability while ensuring Quality of Service (QoS) guarantees. Lotfi and Afghah [2] introduced a meta-learning-based DRL method for dynamic power allocation in Open RAN (O-RAN) environments, demonstrating enhanced adaptability to service variations. Building on this, Rezazadeh

et al. [3] developed TANGO, a graph-based reinforcement learning approach incorporating symbolic reasoning for accurate Physical Resource Block (PRB) allocation in complex 6G topologies. Sohaib et al. [4] extended this direction by proposing energy-aware DRL solutions for balancing eMBB and URLLC requirements under O-RAN scenarios, with meta-learning to enhance generalization. To support flexible and efficient RAN management, Nouruzi et al. [5] proposed a Soft-RAN architecture integrating DRL with Task Offloading Cost (TOC) optimization, targeting reduced overhead and improved network responsiveness. Complementary work by Meignanamoorthi [11] applied customized Advantage Actor-Critic (A2C) algorithms to dynamic sub-slicing, addressing heterogeneous service demands in 6G networks. Zhang et al. [17] explored the fusion of GNN and multi-agent DRL, highlighting its effectiveness in joint resource allocation under varying traffic and service types. Recently, Wang et al. [6] proposed a multi-agent DRL framework with hierarchical GNN embeddings for ultra-dense networks, demonstrating improved scalability and coordination.

Beyond DRL and GNN models, the integration of Software-Defined Networking (SDN) and Multi-access Edge Computing (MEC) has been explored to further enhance decision-making latency and control flexibility. Shi et al. [14] outlined the enabling role of MEC in 6G edge intelligence, while Chen and Liu [13] examined urban V2X networks combining SDN and MEC to optimize vehicular communications. Kreutz et al. [12] surveyed foundational SDN principles applied to 5G and beyond, reinforcing its relevance to dynamic RAN coordination. Additionally, Huang et al. [7] investigated SDN-MEC cooperation for dynamic resource scaling in vehicular edge networks, achieving latency reductions through predictive load balancing.

Several works have also explored advanced radio access techniques for spectral efficiency and interference management. Non-Orthogonal Multiple Access (NOMA) and Rate-Splitting Multiple Access (RSMA) have shown promise in supporting heterogeneous QoS demands [8], [9]. However, most studies treat radio access optimization and task offloading decisions separately, limiting cross-layer synergy. The work by Zhao et al. [10] proposed a cross-layer framework integrating NOMA with edge computing task scheduling, yet without leveraging GNN or DRL for adaptive control.

Despite these advances, existing works often address resource allocation, access control, or task offloading in isolation. In contrast, our proposed iRAN framework integrates SDN-based orchestration, MEC proximity computing, and GNN-enhanced DRL decision-making into a unified, holistic system. This design supports real-time, service-aware, and scalable resource allocation tailored to the stringent and diverse QoS requirements of emerging 6G applications.

Moreover, our model introduces a novel hybrid access scheme combining NOMA and RSMA, allowing the intelligent agent to jointly optimize radio access method selection and resource block allocation per service class. This joint optimization enhances spectral efficiency, user fairness, and la-

tency control, especially in dense and heterogeneous scenarios such as vehicular networks. By simultaneously managing task execution modes (local, SDN-rerouting, MEC offloading) and dynamic access strategies, our iRAN framework achieves superior performance in balancing the trade-offs between latency, reliability, and resource utilization—addressing key limitations of prior works that consider these aspects separately.

This comprehensive integration of multi-layer control, advanced multiple access, and learning-based decision-making constitutes a significant step forward toward realizing practical, efficient, and flexible 6G RAN architectures.

III. MOTIVATION

The next generation of mobile networks, commonly referred to as 6G, is set to revolutionize wireless communications by enabling ultra-reliable and low-latency services, especially in Vehicle-to-Vehicle (V2V) communication scenarios. These services demand strict performance metrics such as ultra-low latency, extreme reliability, and the ability to handle high-density heterogeneous devices [4], [5].

Proximal Policy Optimization (PPO) is a Deep Reinforcement Learning (DRL) method that allows an intelligent agent to make optimal decisions in dynamic and complex environments using stable and efficient policies [1], [15]. This technique is particularly suitable for real-time resource allocation, where network conditions and user demands are continuously evolving.

Graph Neural Networks (GNNs) are machine learning models designed to process structured data in the form of graphs. In the context of V2V networks, where vehicles, base stations (gNodeBs), and MEC servers form an interconnected system, GNNs enable the extraction of spatial and temporal relationships between entities, thereby enhancing system awareness and decision-making [3].

Non-Orthogonal Multiple Access (NOMA) and **Rate-Splitting Multiple Access (RSMA)** are advanced radio access techniques that aim to improve spectral efficiency and interference management in dense environments [8]. These methods allow multiple users to share the same radio resources intelligently, which is critical to meet the increasing capacity demands of 6G networks.

Software-Defined Networking (SDN) provides a flexible and programmable network architecture that separates the control plane from the data plane. In this work, SDN enables dynamic task rerouting among different base stations, thereby optimizing resource utilization and reducing latency [12], [13].

Multi-access Edge Computing (MEC) brings computation capabilities close to end-users, reducing processing delays and relieving the core network load. MEC plays a vital role in executing latency-sensitive computational tasks, such as those encountered in V2V applications [5], [14].

By combining these technologies, this work proposes an integrated framework for intelligent resource allocation that meets the strict requirements of 6G V2V applications while optimizing both networking and computing resources. The primary motivation is to overcome the limitations of traditional,

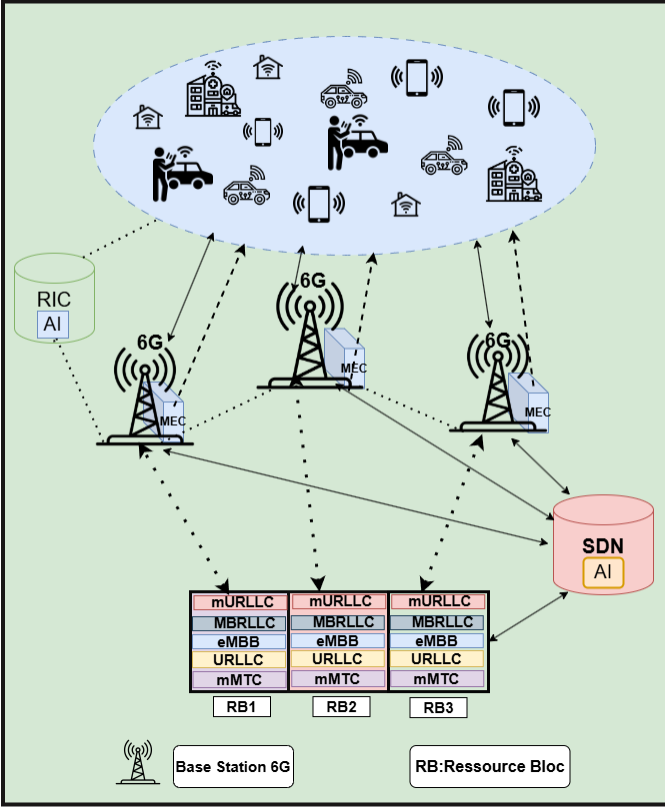


Fig. 1: the architecture of intelligent Radio Access Network (iRAN)

often isolated or static, approaches and to offer a holistic, adaptive, and high-performance solution in the ever-evolving 6G landscape.

IV. PROPOSED APPROACH

A. System Model

Figure 1 illustrates the architecture of the proposed Intelligent Radio Access Network (iRAN), specifically designed to address the stringent requirements and high complexity of 6G networks. The iRAN system is engineered to manage the joint allocation of radio, computing, and storage resources in a highly dynamic and context-aware manner. It supports a wide spectrum of heterogeneous services, including enhanced Mobile Broadband (eMBB), Ultra-Reliable Low Latency Communications (URLLC), and massive Machine-Type Communications (mMTC), as well as their enhanced variants such as mobile URLLC (mURLLC) and mobile broadband Reliable Low Latency Communications (mBRLLC). At the core of the system, User Equipments (UEs) generate computation tasks characterized by a tuple $\{D_i, C_i, \gamma_i, s_i\}$, where D_i denotes the input data size (in bits), C_i the number of required CPU cycles, γ_i the latency constraint, and s_i the service slice type. These delay-sensitive tasks must be handled efficiently to maintain stringent Quality of Service (QoS) and Quality of Experience (QoE) targets.

The iRAN integrates a Multi-access Edge Computing (MEC) layer to provide low-latency computation offloading and local data processing. Radio resources are managed through a hybrid multiple access scheme combining Non-Orthogonal Multiple Access (NOMA) and Rate-Splitting Multiple Access (RSMA), allowing for flexible and spectral-efficient user multiplexing under diverse traffic demands.

To enable intelligent decision-making, the resource allocation process is formulated as a Markov Decision Process (MDP) and solved using a Deep Reinforcement Learning (DRL) framework based on the Proximal Policy Optimization (PPO) algorithm. The network state is encoded into a graph structure, capturing UE-RAN-MEC connectivity, task characteristics, and channel conditions. A Graph Neural Network (GNN) is employed to learn spatial correlations and dynamic interactions among network entities, enabling efficient action selection in large-scale, complex environments.

Furthermore, a Software-Defined Networking (SDN) controller facilitates real-time control and global visibility across distributed RAN nodes. In the case of resource saturation at a gNodeB, the SDN-RIC (RAN Intelligent Controller) orchestrates inter-cell resource reallocation or delegates task execution to nearby MEC servers or cloud nodes.

Overall, the proposed iRAN system empowers the 6G network with intelligent, adaptive, and scalable resource management, while jointly optimizing latency, reliability, spectral efficiency, and energy consumption.

B. iRAN SDN slicing orchestration

The Radio Access Network (RAN) edge consists of gNodeBs, which act as primary access points and local processing units. Upon task arrival, the system attempts to execute the task locally at the serving gNodeB, provided sufficient computation resources are available to meet latency requirements. When local resources are constrained or overloaded, the Software-Defined Networking (SDN) controller reroutes the task to neighboring gNodeBs with available capacity. This centralized coordination optimizes resource utilization and balances the load.

If neither local execution nor inter-gNodeB rerouting can satisfy the task's latency and resource demands due to widespread overload or network congestion the system resorts to offloading the task to Multi-access Edge Computing (MEC) servers. MEC nodes provide enhanced computing power closer to the network edge compared to traditional cloud data centers, thus significantly reducing end-to-end latency.

Overseeing these operations is the RAN Intelligent Controller (RIC), which continuously monitors the network topology, resource availability at gNodeBs and MEC servers, and current traffic conditions. The RIC orchestrates decisions across SDN and MEC components to ensure optimal performance, reliability, and QoS adherence.

To efficiently manage this complex, large-scale, and dynamic environment, the allocation strategy is empowered by Artificial Intelligence (AI). The network is modeled as a graph $G = (V, E)$, where vertices V represent UEs, gNodeBs, and

MEC servers, and edges E represent communication links and interference relationships. A Graph Neural Network (GNN) processes this graph to extract rich spatiotemporal embeddings capturing inter-node dependencies and network state.

These embeddings feed into a Deep Reinforcement Learning (DRL) agent utilizing Proximal Policy Optimization (PPO) to learn an optimal policy for task execution decisions. The agent dynamically selects among multiple actions local UE processing, local gNodeB execution, SDN-based rerouting to neighboring gNodeBs, or offloading to MEC servers with the objective to minimize overall task latency while satisfying heterogeneous QoS requirements associated with diverse 6G service classes.

This integrated framework enables scalable, adaptive, and service-aware resource management aligned with the stringent demands of next-generation wireless systems.

a) Resource Allocation Schemes

The proposed iRAN framework (Fig 2) incorporates an intelligent and hierarchical resource allocation strategy that adapts in real time to dynamic network conditions, user demands, and heterogeneous service requirements.

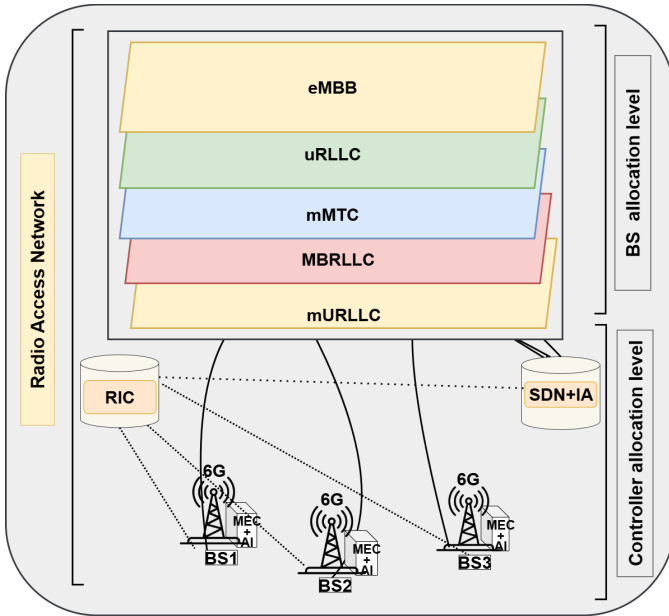


Fig. 2: iRAN resource allocation framework integrating NOMA, RSMA, MEC, DRL-PPO, GNN, and SDN-RIC

It jointly considers task offloading and radio access technique selection to guarantee efficient utilization of computational and spectral resources while satisfying stringent Quality of Service (QoS) constraints. The decision-making process, orchestrated by the RAN Intelligent Controller (RIC), encompasses the following three main execution modes:

- 1) **Local gNodeB Execution with Hybrid NOMA/RSMA:** When the serving gNodeB has sufficient computing and radio resources, tasks are processed locally to minimize latency. The radio

access employs a hybrid multiple access scheme combining Non-Orthogonal Multiple Access (NOMA) and Rate-Splitting Multiple Access (RSMA), enabling better spectrum reuse and interference management. This hybrid access technique supports simultaneous transmissions to multiple users with diverse QoS profiles, including eMBB, URLLC, and mMTC services. By dynamically adapting the power allocation (for NOMA) and the message splitting ratios (for RSMA), the system enhances spectral efficiency while maintaining reliability and low delay.

- 2) **Inter-gNodeB Task Rerouting via SDN-based Coordination:** In cases where the local gNodeB cannot fulfill the task requirements due to resource scarcity or suboptimal radio conditions, the Software-Defined Networking (SDN) controller initiates a rerouting mechanism. Tasks are redirected to adjacent gNodeBs with surplus capacity. This process is coupled with a dynamic reassessment of the optimal access strategy (NOMA, RSMA, or hybrid) based on the load distribution and service type at the destination gNodeB. The SDN controller thus enables fine-grained load balancing and latency-aware resource delegation across the RAN.
- 3) **MEC Offloading with Flexible Access Adaptation:** If neither local execution nor inter-gNodeB rerouting can meet the latency and QoS requirements, tasks are offloaded to nearby Multi-access Edge Computing (MEC) servers. These MEC nodes provide high-performance computing resources at the network edge, ensuring compliance with critical time constraints. The access mechanism at this level remains flexible, switching adaptively between NOMA and RSMA depending on user density, interference levels, and service urgency. This multi-level adaptation significantly improves task completion probability and system responsiveness under highly dynamic conditions.

The RAN Intelligent Controller (RIC) plays a central role in orchestrating these decisions. It leverages real-time traffic analytics, historical usage data, and machine learning-based prediction models to continuously optimize task placement and access mode selection. By jointly optimizing computation offloading, gNodeB cooperation, and hybrid multiple access strategies, the iRAN framework provides an intelligent, resilient, and scalable resource allocation mechanism tailored for next-generation 6G networks.

C. Analytical Study

In this section, we present the analytical foundations of the proposed iRAN framework, which integrates heterogeneous service requirements, multi-mode task execution, and intelligent control through Graph Neural Networks (GNN) and Proximal Policy Optimization (PPO). The model captures the trade-offs between latency, energy, and Quality of Service (QoS) compliance for real-time decision-making in 6G Radio Access Networks.

1) *Task Characterization and Latency Constraints*: Each User Equipment (UE) i periodically generates computation-intensive tasks defined by the tuple $(D_i, C_i, \gamma_i, R_i, s_i)$, where:

- D_i (in bits): Size of the input data to be processed.
- C_i (in cycles/bit): Number of CPU cycles required per bit for execution.
- γ_i (in ms): Maximum tolerable latency for the task to satisfy QoS.
- R_i (in bps): Achievable data rate between the UE and its serving node.
- s_i : Service type (or slice), where $s_i \in \{\text{eMBB, URLLC, mMTC, mURLLC, mbRLLC}\}$.

Latency constraints are service-specific and defined according to 6G standard guidelines [4], [5], [13], [14] as follows:

$$\gamma_i = \begin{cases} 1 & \text{if } s_i = \text{URLLC}, \\ 5 & \text{if } s_i = \text{mURLLC}, \\ 20 & \text{if } s_i = \text{mbRLLC}, \\ 100 & \text{if } s_i = \text{eMBB}, \\ 500 & \text{if } s_i = \text{mMTC}. \end{cases} \quad (1)$$

2) *Energy Consumption Model*: The energy consumed by UE i under execution mode x_i is modeled using a dynamic CPU frequency scaling approach:

$$E_i(x_i) = \kappa \cdot C_i \cdot D_i \cdot f_i^2(x_i) \quad (2)$$

where:

- κ is the effective switched capacitance of the processor,
- $f_i(x_i)$ is the CPU frequency used under the selected mode x_i ,
- $x_i \in \{0, 1, 2, 3\}$ indicates the execution location (local, gNodeB, rerouted, MEC).

This quadratic model reflects the energy-performance trade-off where higher CPU frequencies enable lower latency but incur greater energy cost.

3) *Task Execution Latency per Mode*: The task latency $T_i(x_i)$ varies with execution mode:

a) *Local execution at gNodeB* ($x_i = 1$):

$$T_i^{(1)} = \frac{D_i}{R_i} + \varepsilon, \quad \varepsilon \ll 1 \quad (3)$$

Here, ε accounts for minimal queueing or processing delay at the gNodeB.

b) *SDN-based rerouting to another gNodeB* ($x_i = 2$):

$$T_i^{(2)} = \frac{D_i}{R_i} + \delta + \frac{D_i}{R'_i} + \frac{D_i C_i}{f_j} \quad (4)$$

with:

- δ : signaling and SDN rerouting overhead (ms),
- R'_i : achievable rate to the target gNodeB,
- f_j : processing frequency at the destination gNodeB.

c) *Local execution on UE* ($x_i = 0$):

$$T_i^{\text{local}} = \frac{D_i \cdot C_i}{f_i} \quad (5)$$

d) *MEC execution* ($x_i = 3$):

$$T_i^{\text{MEC}} = \frac{D_i}{R_i} + \frac{D_i \cdot C_i}{F_i^{\text{MEC}}} \quad (6)$$

where F_i^{MEC} is the MEC server's computing capability allocated to task i .

4) *Optimal Execution Decision*: We define the optimal execution decision x_i^* as:

$$\begin{aligned} x_i^* &= \arg \min_{x_i \in \{0, 1, 2, 3\}} T_i(x_i) \\ \text{s.t. } & T_i(x_i) \leq \gamma_i, \\ & R_i \geq \delta_i, \\ & P_i^{\text{success}} \geq \eta_i \end{aligned} \quad (7)$$

where:

- δ_i : minimum data rate requirement for service s_i ,
- η_i : reliability threshold (e.g., target packet success rate).

5) *AI-Based Control: GNN-Driven PPO Policy*: The environment is represented as a dynamic graph $G = (V, E)$ where:

- Each node $v_i \in V$ represents a UE, gNodeB, or MEC server,
- Each edge $(i, j) \in E$ encodes radio connectivity or interference relations.

Each node has a feature vector:

$$s_i = [D_i, C_i, f_i, R_i, \gamma_i, \text{enc}(s_i)]$$

The GNN extracts spatiotemporal features and produces embeddings used by a Proximal Policy Optimization (PPO) agent. The action space is:

- $x_i = 0$: Local UE execution,
- $x_i = 1$: gNodeB processing,
- $x_i = 2$: SDN-based rerouting,
- $x_i = 3$: MEC offloading.

6) *Reward and Utility Modeling*: The PPO agent is trained using a reward function designed to capture latency minimization, QoS satisfaction, and energy efficiency:

$$R_i = -T_i(x_i) + \lambda_1 \cdot QoS_i - \lambda_2 \cdot E_i(x_i) \quad (8)$$

where:

- $QoS_i = \mathbb{I}_{\{T_i(x_i) \leq \gamma_i\}}$: binary indicator of QoS success,
- λ_1, λ_2 : tunable weights balancing performance goals.

To further reflect service heterogeneity, we define a utility function:

$$\mathcal{U}_i = w_{\text{lat}}^{(s_i)} \cdot \alpha_i^{QoS} - w_E^{(s_i)} \cdot E_i(x_i) \quad (9)$$

with:

- $\alpha_i^{QoS} = 1$ if latency constraint is satisfied; 0 otherwise,
- $w_{\text{lat}}^{(s_i)}, w_E^{(s_i)}$: service-dependent weights for latency sensitivity and energy importance.

A class-aware reward function is given by:

$$\mathcal{R}_i = -T_i(x_i) \cdot w_{\text{lat}}^{(s_i)} + \beta_{s_i} \cdot \alpha_i^{QoS} - E_i(x_i) \cdot w_E^{(s_i)} \quad (10)$$

where β_{s_i} represents the QoS bonus associated with service slice s_i .

7) *SDN-Enabled Global Coordination*: The Software-Defined Networking (SDN) controller centrally monitors network state through resource availability vectors:

$$b_j = [c_j^{\text{CPU}}, m_j^{\text{MEM}}, r_j^{\text{RADIO}}]$$

for each gNodeB j , where each component reflects normalized availability of CPU cycles, memory, and radio bandwidth. Upon congestion detection, the controller queries neighboring nodes and selects the rerouting path that minimizes joint latency and energy. The RAN Intelligent Controller (RIC) further ensures policy consistency across distributed agents and updates global PPO parameters periodically through federated learning or asynchronous feedback mechanisms.

D. Proposed Algorithm

Algorithm 1 iRAN Resource Allocation with DRL and GNN

Input: User Equipments set $\mathcal{U}$, gNodeBs set $\mathcal{G}$, SDN resource matrix B , service classes $\{s_i\}$, task parameters $\{D_i, C_i, f_i, \gamma_i\}$, MEC capacities $\{F_j^{\text{MEC}}\}$

Output: Optimal execution actions $\{x_i\}$ for each UE_i

foreach $UE_i \in \mathcal{U}$ **do**

 // Step 1: Observe task and network state

$$s_i = (D_i, C_i, f_i, R_i, \gamma_i, s_i^{\text{class}}, \text{neighbor states})$$

 // Step 2: Construct dynamic network graph

 Build graph $\mathcal{GN} = (V, E)$ where:

- V : nodes representing UEs, gNodeBs, and MEC servers
- E : edges encoding communication links, interference, and SDN routing relations

 // Step 3: Extract contextual embeddings

$$\mathbf{h}_i = \text{GNN_Encoder}(s_i, \mathcal{GN})$$

 // Step 4: Select action via PPO policy

$$x_i \sim \pi_\theta(\cdot | \mathbf{h}_i), \quad x_i \in \begin{cases} 0: & \text{UE local,} \\ 1: & \text{local gNodeB,} \\ 2: & \text{SDN rerouting,} \\ 3: & \text{MEC offloading} \end{cases}$$

 // Step 5: Handle SDN rerouting

if $x_i = 2$ **then**

 Query SDN controller with B to find alternative gNodeB g_j
 with adequate resources Reroute task to g_j

 // Step 6: Compute latency for selected mode

$$T_i^{\text{UE}} = \frac{D_i C_i}{f_i}$$

$$T_i^{\text{tx}} = \frac{D_i}{R_i}$$

$$T_i^{\text{gNodeB}} = T_i^{\text{tx}} + \varepsilon \quad (\varepsilon : \text{processing delay})$$

$$T_i^{\text{SDN}} = T_i^{\text{tx}} + \delta + \frac{D_i}{R'_i} + \frac{D_i C_i}{f_j}$$

$$T_i^{\text{MEC}} = T_i^{\text{tx}} + \frac{D_i C_i}{F_j^{\text{MEC}}}$$

 // Step 7: Calculate reward

 Evaluate energy consumption $E_i(x_i)$ (cf. Eq. (2)) and QoS indicator $QoS(s_i)$, then:

$$R_i = -T_i(x_i) + \lambda_1 \cdot QoS(s_i) - \lambda_2 \cdot E_i(x_i)$$

 // Step 8: Update PPO policy

 Update policy parameters θ via gradient ascent:

$$\theta \leftarrow \theta + \alpha \nabla_\theta \mathbb{E}_{x_i \sim \pi_\theta} [R_i]$$

return Optimized action set $\{x_i\}$

To address the rising complexity of heterogeneous services and dynamic network conditions in 6G iRAN systems, we propose an intelligent resource allocation algorithm that integrates Deep Reinforcement Learning (DRL) with Graph Neural Networks (GNN).

This method dynamically selects the optimal execution mode for each user task — either processed locally at the serving gNodeB, rerouted via SDN to neighboring cells, or offloaded to MEC — while jointly optimizing latency, energy consumption, and QoS compliance.

Key Features of the Algorithm:

- *Observation and Contextual Embedding*: Each UE’s task characteristics and environment context, including neighboring nodes’ states and resource availability, are encoded as graph node features. The GNN captures complex spatial-temporal dependencies such as interference patterns, resource contention, and network topology changes.
- *Decision-Making via PPO*: Leveraging the rich node embeddings, the PPO-based DRL agent learns a stochastic policy to balance latency reduction, energy efficiency, and QoS adherence, adapting continuously to varying network dynamics and heterogeneous service demands.
- *SDN-Assisted Rerouting*: When local resources are insufficient, the algorithm utilizes the SDN controller’s global network view to identify alternative gNodeBs with available capacity, thus mitigating congestion and latency issues.
- *MEC Offloading for QoS Assurance*: MEC nodes provide powerful edge computing resources, enabling offloading when local execution and rerouting cannot satisfy strict latency and reliability constraints.
- *Multi-Objective Reward Design*: The reward function is carefully designed to jointly optimize latency minimization, QoS satisfaction, and energy consumption, ensuring a balanced and robust resource allocation strategy.

This algorithmic framework supports real-time, scalable, and service-aware resource allocation in iRANs, paving the way for efficient and adaptive management of next-generation 6G networks with diverse and stringent application requirements.

V. PERFORMANCE EVALUATION

A. Simulation Environment

The simulation was conducted on Google Colab using Python-based tools to model and evaluate the iRAN system. The environment considers 6G scenarios involving heterogeneous services, user mobility, and dynamic network conditions. Representative use cases such as URLLC in vehicle-to-vehicle (V2V) communications were considered to assess the system’s responsiveness under strict latency constraints.

Training was performed over 100 episodes, and key metrics such as task latency, energy consumption, reward, and QoS

satisfaction were monitored. The environment can also support the evaluation of other use cases such as eMBB (e.g., high-definition video streaming), offering a broader perspective on the framework’s adaptability.

B. Simulation Parameters

We evaluate the proposed iRAN framework in a simulated urban 6G Vehicle-to-Vehicle (V2V) scenario comprising 10 gNodeBs and 10 MEC servers. The infrastructure also includes two SDN controllers for dynamic routing and task offloading, and one RAN Intelligent Controller (RIC) responsible for centralized orchestration and policy enforcement [12]–[14].

The network supports diverse 6G service types such as eMBB, URLLC, mMTC, as well as enhanced variants like mURLLC and mBRLLC. Traffic demands vary over time and space due to vehicular mobility and dynamic load fluctuations. The simulation spans multiple episodes to assess key performance indicators, including average task latency, cumulative agent reward, energy consumption, and Quality of Service (QoS) satisfaction across different resource allocation strategies.

TABLE I: Key Simulation Parameters

Parameter	Value
Number of gNodeBs	10
Number of MEC Servers	10
Number of SDN Controllers	2
Number of RICs	1
Bandwidth	20 MHz
Access Scheme	Hybrid NOMA / RSMA
Mobility Model	Urban V2V
Simulation Episodes	100

C. Results and Interpretation

The integration of Graph Neural Networks (GNN) with Proximal Policy Optimization (PPO) significantly enhances resource allocation performance in 6G Vehicle-to-Vehicle (V2V) communication scenarios. By explicitly modeling the network’s intrinsic graph-structured topology and capturing complex spatial-temporal dependencies among nodes—such as interference patterns, mobility dynamics, and heterogeneous traffic demands—the PPO+GNN approach achieves a substantial reduction in end-to-end latency. Specifically, the average latency decreases from 17.63 ms with the baseline PPO model to 7.68 ms, representing a remarkable improvement of approximately 56% (see Fig. 3). This reduction is critical for safety-critical V2V applications, where ultra-low delay and high reliability are mandatory to ensure timely information exchange, collision avoidance, and cooperative maneuvers.

Beyond latency improvements, the PPO+GNN model demonstrates superior overall efficiency, as reflected by its higher cumulative reward of 25.06 compared to 14.82 for the baseline PPO agent (Fig. 4). This reward metric encapsulates the model’s balanced optimization across multiple conflicting objectives, including latency minimization, Quality of Service (QoS) adherence, and network load management. The

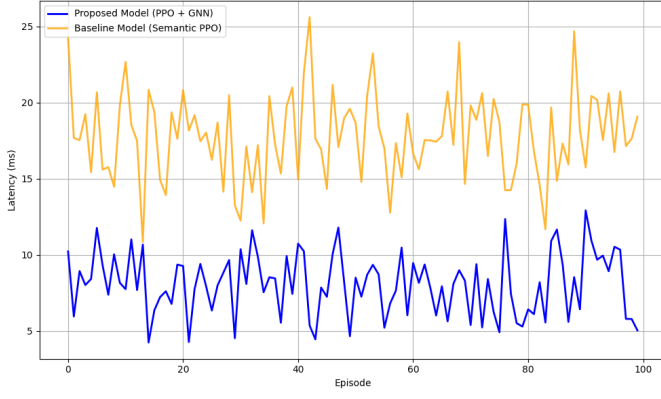


Fig. 3: Latency Comparison between baseline PPO and PPO+GNN

enhanced reward indicates that PPO+GNN not only reduces delays but also better manages network resources to maintain service-level agreements and user satisfaction.

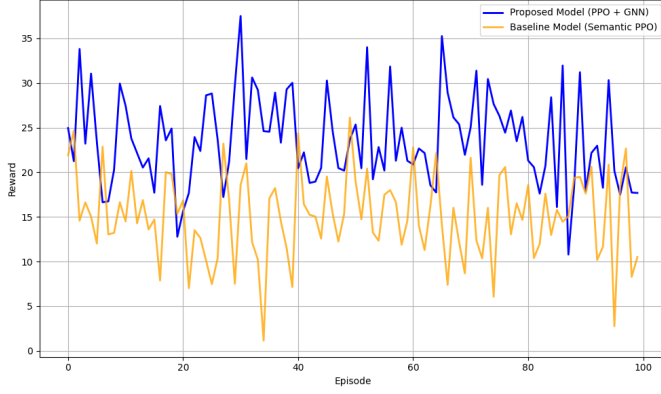


Fig. 4: Cumulative Reward Comparison

Correspondingly, the QoS satisfaction—defined as the percentage of tasks that meet their stringent latency and reliability constraints—improves from 64.94% under the baseline PPO to 67.71% with the PPO+GNN model (Fig. 5). This incremental gain, although seemingly modest, is significant in the context of URLLC and other latency-critical services, where even small increases in compliance can translate into markedly better user experience and system reliability. The model’s ability to accommodate diverse 6G service requirements, spanning URLLC, enhanced Mobile Broadband (eMBB), and emerging classes like mURLLC and mbRLLC, underscores its adaptability and robustness in heterogeneous traffic environments.

The improved performance is achieved with a moderate increase in energy consumption, rising from 1.99 to 2.47 energy units as shown in Fig. 6. This increment primarily stems from two factors: the additional computational overhead introduced by GNN processing and the higher frequency of task offloading to Multi-access Edge Computing (MEC) nodes, which consume more energy compared to local processing at UEs or gNodeBs. Despite this, the trade-off remains favorable

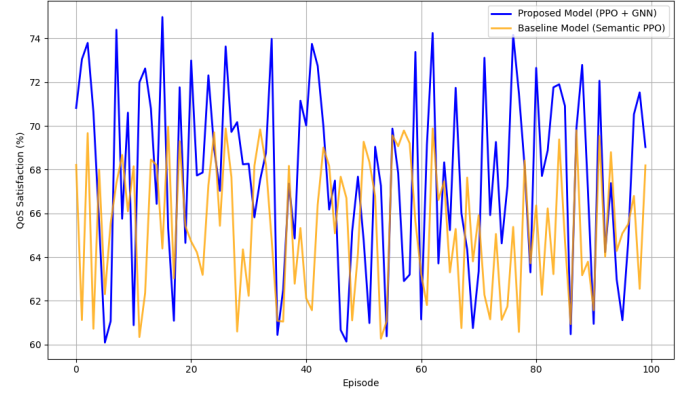


Fig. 5: Quality of Service (QoS) Satisfaction Comparison

because energy increase is outweighed by substantial gains in latency reduction and QoS satisfaction, which are paramount in mission-critical vehicular networks. Future work may explore energy-efficient GNN architectures or offloading policies to further mitigate this overhead.

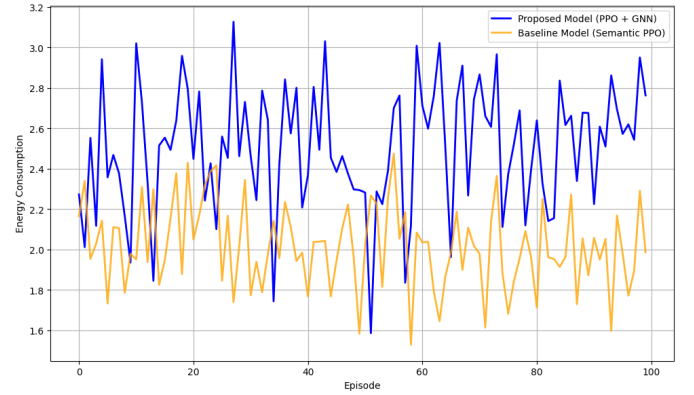


Fig. 6: Energy Consumption Comparison

TABLE II: Performance Comparison between Baseline PPO and PPO+GNN Models

Metric	Baseline PPO	PPO+GNN
Average Latency (ms)	17.63	7.68
Cumulative Reward	14.82	25.06
QoS Satisfaction (%)	64.94	67.71
Energy Consumption (units)	1.99	2.47

Overall, these results highlight the efficacy of embedding graph-based representations within deep reinforcement learning frameworks to capture the complex spatial and temporal correlations inherent to dynamic V2V networks. By leveraging these rich embeddings, the DRL agent can adapt resource allocation policies more effectively in response to fluctuating traffic loads, highly mobile user behaviors, and heterogeneous service demands. Such adaptability is essential to meet the ultra-reliable low-latency communication (URLLC) and en-

hanced mobile broadband (eMBB) requirements expected in future 6G deployments.

Furthermore, the graph-informed DRL approach exhibits promising scalability potential. Due to its inherent capacity to abstract and aggregate information via graph structures, the PPO+GNN method can generalize to larger and denser vehicular networks with minimal performance degradation. This scalability is critical for supporting the evolution toward highly connected and intelligent transportation systems, where dense V2V interactions, massive sensor data, and diverse service requirements coexist.

In summary, the experimental findings demonstrate that the proposed PPO+GNN framework offers a powerful, flexible, and efficient solution for intelligent resource management in 6G V2V communication, paving the way for safer, faster, and more reliable vehicular networks.

VI. CONCLUSION AND FUTURE WORK

This paper has presented a comprehensive analytical study and performance evaluation of an intelligent resource allocation framework specifically designed for 6G Vehicle-to-Vehicle (V2V) Ultra-Reliable Low-Latency Communications (URLLC). By leveraging Proximal Policy Optimization (PPO) integrated with Graph Neural Networks (GNN), the proposed model effectively captures the complex, dynamic network topology and heterogeneous service requirements characteristic of emerging 6G networks.

Extensive simulation results demonstrate that the PPO+GNN approach significantly outperforms baseline PPO models across key performance metrics, including substantial latency reduction and improved Quality of Service (QoS) satisfaction rates, while maintaining a favorable trade-off with energy consumption. The integration of GNNs enables context-aware and topology-sensitive decision-making by encoding relational and temporal dependencies among network entities, which is critical to overcoming challenges posed by high mobility, interference dynamics, and fluctuating resource availability in V2V scenarios.

This work validates the potential of combining advanced deep reinforcement learning techniques with graph-based representations to optimize complex, multi-agent resource allocation problems in next-generation wireless networks. The adaptive and scalable nature of the proposed framework positions it as a promising candidate for deployment in real-world intelligent transportation systems, where heterogeneous services and stringent performance guarantees coexist.

Looking forward, several promising research avenues remain to be explored. Incorporating more realistic and comprehensive mobility models, including varying traffic densities and vehicle trajectories, as well as communication impairments such as channel fading, shadowing, interference, and packet loss, will increase the fidelity and robustness of the resource allocation model. Furthermore, investigating decentralized and federated multi-agent reinforcement learning frameworks could enhance scalability, fault tolerance, and privacy preservation in ultra-dense vehicular deployments by

enabling distributed learning and decision-making without relying on a centralized controller.

Security and privacy represent critical dimensions for future V2V systems. Integrating secure and privacy-preserving mechanisms directly within the resource allocation framework—such as secure multi-party computation, differential privacy, or blockchain-inspired trust management—will be essential to guarantee reliable and trustworthy communications in the face of increasing cyber threats.

Moreover, the advent of quantum technologies opens exciting new frontiers. Quantum Neural Networks (QNNs) and quantum-enhanced reinforcement learning algorithms possess the potential to efficiently address the curse of dimensionality inherent in large-scale optimization problems, offering accelerated learning and decision-making capabilities. The integration of Quantum Key Distribution (QKD) could ensure provably secure communication channels, thus addressing the escalating cybersecurity concerns in connected vehicular networks. Exploring hybrid quantum-classical architectures tailored for 6G V2V systems thus constitutes a compelling research direction, potentially unlocking unprecedented levels of efficiency, security, and adaptability in future intelligent transportation systems.

In conclusion, this work lays a solid foundation for intelligent, adaptive, and secure resource management in 6G V2V networks and highlights promising pathways for integrating emerging AI and quantum technologies to meet the stringent demands of next-generation vehicular communications.

DECLARATIONS

All authors declare that they have no conflicts of interest.

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